Pepcoding DSA 4Month Study Plan

Bachelor's of technology (Rayat Bahra University)

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PEPCODING

DSA Level 1 & Level 2

Complete Study Plan & Question Bank

4-Month Structured Roadmap

Day-wise • Week-wise • Month-wise

Approx. 200+ Questions | 4 Months | ~2 Questions/Day

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4-Month Plan Overview

This document covers all DSA topics from Pepcoding Level 1 & Level 2 playlists, organized into a structured 4-month plan. Each day you solve approximately 2 questions, progressing from basics to advanced concepts.

Month	Focus Area	Weeks	Questions
Month 1	Arrays, Recursion, Strings, Maths, Bit Manipulation	W1–W4	~50
Month 2	Linked List, Stack, Queue, Hashing, Binary Search	W5–W8	~48
Month 3	Trees, BST, Heaps, Graphs (Basic + Advanced)	W9–W12	~55
Month 4	Dynamic Programming, Tries, Segment Trees, Greedy, Sliding Window	W13–W16	~55

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□ Level 1 — Question Bank

All questions extracted from Pepcoding DSA Level 1 Playlist, organized topic-wise in order of difficulty.

#	Topic	Questions
1	**Introduction & Time-Space Complexity**	Big O Notation Basics
		Time Complexity Analysis
		Space Complexity Analysis
		Best / Worst / Average Case
		Asymptotic Notations
2	**Arrays - Basics**	Introduction to Arrays
		Linear Search
		Binary Search
		Bubble Sort
		Selection Sort
		Insertion Sort
		Find Max/Min in Array
		Reverse an Array
		Check Sorted Array
3	**Arrays - Intermediate**	Find Duplicate Element
		Rotate Array by K
		Move Zeros to End
		Two Sum (Unsorted)
		Two Sum (Sorted)
		Pair Sum Closest to Target
		Triplet Sum
		Subarray with Zero Sum
		Subarray Sum Equals K
4	**Recursion & Backtracking**	Factorial using Recursion
		Fibonacci using Recursion
		Power(x, n)
		Print Numbers 1 to N
		Reverse a String using Recursion
		Check Palindrome using Recursion

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		Tower of Hanoi
		Print All Subsequences
		Print All Permutations of String
		N-Queens Problem
		Rat in a Maze
		Sudoku Solver
5	**Strings**	Reverse a String
		Check Palindrome
		Count Vowels & Consonants
		Remove Duplicates from String
		Anagram Check
		Longest Common Prefix
		Count and Say
		Roman to Integer
		Integer to Roman
		Valid Parentheses
		Longest Palindromic Substring (Brute Force)
6	**Maths for DSA**	Check Prime Number
		Sieve of Eratosthenes
		GCD & LCM
		Count Digits
		Reverse a Number
		Check Armstrong Number
		Sum of Digits
		Count Set Bits
		Power of 2
7	**Bit Manipulation**	Check if Kth Bit Set
		Set Kth Bit
		Clear Kth Bit
		Toggle Kth Bit
		Count Set Bits (Brian Kernighan)
		Find Single Non-Duplicate
		XOR of All Subsets

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8	**Linked List**	Introduction & Node Creation
		Insert at Beginning / End / Position
		Delete at Beginning / End / Position
		Search in Linked List
		Reverse a Linked List
		Detect Loop (Floyd's Algorithm)
		Remove Loop
		Find Middle Element
		Merge Two Sorted Lists
		Check Palindrome Linked List
		Intersection Point of Two Lists
9	**Stack**	Stack using Array
		Stack using Linked List
		Valid Parentheses (Stack)
		Next Greater Element
		Next Smaller Element
		Stock Span Problem
		Min Stack
		Evaluate Postfix Expression
		Infix to Postfix
10	**Queue**	Queue using Array
		Queue using Linked List
		Queue using Two Stacks
		Circular Queue
		Deque (Double Ended Queue)
		Sliding Window Maximum
		First Non-Repeating Character in Stream
11	**Binary Search (Advanced)**	Search in Rotated Sorted Array
		Find Peak Element
		Median of Two Sorted Arrays (Brute)
		Kth Smallest / Largest
		Allocate Books (Binary Search on Answer)
		Painter's Partition

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		Aggressive Cows
		Capacity to Ship Packages
12	**Hashing**	HashMap Basics
		Frequency Count
		Two Sum using Hashing
		Longest Consecutive Sequence
		Group Anagrams
		Subarray Sum Equals K (Hashing)
		Top K Frequent Elements
		Intersection & Union of Arrays

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□ Level 2 — Question Bank

All questions extracted from Pepcoding DSA Level 2 Playlist, organized topic-wise in order of difficulty.

#	Topic	Questions
1	**Trees - Basics**	Binary Tree Introduction & Node Creation
		Tree Traversals (Inorder, Preorder, Postorder)
		Level Order Traversal (BFS)
		Height of Binary Tree
		Count Nodes
		Sum of All Nodes
		Check Balanced Tree
		Diameter of Binary Tree
		Mirror / Invert Binary Tree
		Path from Root to Node
		Lowest Common Ancestor
2	**Binary Search Tree (BST)**	Search in BST
		Insert in BST
		Delete in BST
		Kth Smallest in BST
		Validate BST
		Inorder Successor / Predecessor
		Convert Sorted Array to BST
		BST to Greater Sum Tree
3	**Heaps / Priority Queue**	Min Heap & Max Heap
		Heapify
		Heap Sort
		Kth Largest Element
		Top K Frequent Elements (Heap)
		Merge K Sorted Lists
		Find Median from Data Stream
4	**Graphs - Basics**	Graph Representation (Adjacency List & Matrix)
		BFS on Graph
		DFS on Graph

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		Detect Cycle (Undirected - BFS/DFS)
		Detect Cycle (Directed - DFS)
		Topological Sort (BFS - Kahn's)
		Topological Sort (DFS)
		Bipartite Check
		Number of Islands
		Flood Fill
5	**Graphs - Advanced**	Dijkstra's Algorithm
		Bellman-Ford Algorithm
		Floyd-Warshall Algorithm
		Prim's Algorithm (MST)
		Kruskal's Algorithm (MST)
		Bridges in Graph
		Articulation Points
		Strongly Connected Components (Kosaraju)
6	**Dynamic Programming - 1D**	Fibonacci (Memoization & Tabulation)
		Climbing Stairs
		House Robber
		House Robber II (Circular)
		Jump Game
		Jump Game II (Min Jumps)
		Decode Ways
		Minimum Cost Climbing Stairs
		Partition Equal Subset Sum
7	**Dynamic Programming - 2D (Grid)**	Unique Paths
		Unique Paths II (Obstacles)
		Minimum Path Sum
		Triangle (Min Path)
		Maximal Square
		Dungeon Game
		Cherry Pickup
8	**DP - Subsequences & Substrings**	Longest Common Subsequence (LCS)
		Edit Distance

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		Longest Increasing Subsequence
		Longest Common Substring
		Shortest Common Supersequence
		Count Distinct Subsequences
		Wildcard Matching
		Regular Expression Matching
9	**DP - Knapsack Variants**	0/1 Knapsack
		Unbounded Knapsack
		Fractional Knapsack (Greedy)
		Rod Cutting
		Coin Change (Min Coins)
		Coin Change II (Count Ways)
		Target Sum (DP)
		Boolean Parenthesization
10	**Tries**	Trie Insert & Search
		Prefix Search / Auto-Complete
		Count Words with Prefix
		Longest Common Prefix (Trie)
		Word Break Problem (Trie)
		Maximum XOR of Two Numbers
11	**Segment Trees & Advanced DS**	Segment Tree - Build
		Segment Tree - Range Query
		Segment Tree - Point Update
		Lazy Propagation
		Fenwick Tree (BIT)
		Sparse Table (Range Min/Max)
12	**Greedy Algorithms**	Activity Selection Problem
		Fractional Knapsack
		Job Sequencing with Deadlines
		Huffman Encoding
		N Meetings in One Room
		Minimum Platforms
		Candy Distribution

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		Gas Station
13	**Sliding Window & Two Pointers**	Maximum Sum Subarray of Size K
		Longest Substring Without Repeating Chars
		Minimum Window Substring
		Longest Substring with At Most K Distinct
		Count Subarrays with Product Less than K
		Container with Most Water
		Trapping Rain Water
		4-Sum Problem

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□ Month-wise Study Plan

A detailed day-by-day breakdown for all 4 months. Complete each day's questions before moving to the next.

Month 1 — Overview

Focus: Arrays · Recursion · Strings · Maths · Bit Manipulation

Day	Topic	Questions
Day 1	Introduction & Time-Space Complexity	Big O Notation Basics / Time Complexity Analysis
Day 2	Introduction & Time-Space Complexity	Space Complexity Analysis / Best / Worst / Average Case
Day 3	Introduction & Time-Space Complexity, Arrays - Basics	Asymptotic Notations / Introduction to Arrays
Day 4	Arrays - Basics	Linear Search / Binary Search
Day 5	Arrays - Basics	Bubble Sort / Selection Sort
Day 6	Arrays - Basics	Insertion Sort / Find Max/Min in Array
Day 7	Arrays - Basics	Reverse an Array / Check Sorted Array
Day 8	Arrays - Intermediate	Find Duplicate Element / Rotate Array by K
Day 9	Arrays - Intermediate	Move Zeros to End / Two Sum (Unsorted)
Day 10	Arrays - Intermediate	Two Sum (Sorted) / Pair Sum Closest to Target
Day 11	Arrays - Intermediate	Triplet Sum / Subarray with Zero Sum
Day 12	Arrays - Intermediate, Recursion & Backtracking	Subarray Sum Equals K / Factorial using Recursion
Day 13	Recursion & Backtracking	Fibonacci using Recursion / Power(x, n)
Day 14	Recursion & Backtracking	Print Numbers 1 to N / Reverse a String using Recursion
Day 15	Recursion & Backtracking	Check Palindrome using Recursion / Tower of Hanoi
Day 16	Recursion & Backtracking	Print All Subsequences / Print All Permutations of String
Day 17	Recursion & Backtracking	N-Queens Problem / Rat in a Maze
Day 18	Recursion & Backtracking, Strings	Sudoku Solver / Reverse a String
Day 19	Strings	Check Palindrome / Count Vowels & Consonants
Day 20	Strings	Remove Duplicates from String / Anagram Check
Day 21	Strings	Longest Common Prefix / Count and Say
Day 22	Strings	Roman to Integer / Integer to Roman
Day 23	Strings	Valid Parentheses / Longest Palindromic Substring (Brute Force)
Day 24	Maths for DSA	Check Prime Number / Sieve of Eratosthenes
Day 25	Maths for DSA	GCD & LCM / Count Digits

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Day 26	Maths for DSA	Reverse a Number / Check Armstrong Number
Day 27	Maths for DSA	Sum of Digits / Count Set Bits
Day 28	Maths for DSA, Bit Manipulation	Power of 2 / Check if Kth Bit Set
Day 29	Bit Manipulation	Set Kth Bit / Clear Kth Bit
Day 30	Bit Manipulation	Toggle Kth Bit / Count Set Bits (Brian Kernighan)

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Month 2 — Overview

Focus: Linked List · Stack · Queue · Hashing · Binary Search

Day	Topic	Questions
Day 31	Bit Manipulation	Find Single Non-Duplicate / XOR of All Subsets
Day 32	Linked List	Introduction & Node Creation / Insert at Beginning / End / Position
Day 33	Linked List	Delete at Beginning / End / Position / Search in Linked List
Day 34	Linked List	Reverse a Linked List / Detect Loop (Floyd's Algorithm)
Day 35	Linked List	Remove Loop / Find Middle Element
Day 36	Linked List	Merge Two Sorted Lists / Check Palindrome Linked List
Day 37	Linked List, Stack	Intersection Point of Two Lists / Stack using Array
Day 38	Stack	Stack using Linked List / Valid Parentheses (Stack)
Day 39	Stack	Next Greater Element / Next Smaller Element
Day 40	Stack	Stock Span Problem / Min Stack
Day 41	Stack	Evaluate Postfix Expression / Infix to Postfix
Day 42	Queue	Queue using Array / Queue using Linked List
Day 43	Queue	Queue using Two Stacks / Circular Queue
Day 44	Queue	Deque (Double Ended Queue) / Sliding Window Maximum
Day 45	Queue, Binary Search (Advanced)	First Non-Repeating Character in Stream / Search in Rotated Sorted Array
Day 46	Binary Search (Advanced)	Find Peak Element / Median of Two Sorted Arrays (Brute)
Day 47	Binary Search (Advanced)	Kth Smallest / Largest / Allocate Books (Binary Search on Answer)
Day 48	Binary Search (Advanced)	Painter's Partition / Aggressive Cows
Day 49	Binary Search (Advanced), Hashing	Capacity to Ship Packages / HashMap Basics
Day 50	Hashing	Frequency Count / Two Sum using Hashing
Day 51	Hashing	Longest Consecutive Sequence / Group Anagrams
Day 52	Hashing	Subarray Sum Equals K (Hashing) / Top K Frequent Elements
Day 53	Hashing, Trees - Basics	Intersection & Union of Arrays / Binary Tree Introduction & Node Creation
Day 54	Trees - Basics	Tree Traversals (Inorder, Preorder, Postorder) / Level Order Traversal (BFS)
Day 55	Trees - Basics	Height of Binary Tree / Count Nodes
Day 56	Trees - Basics	Sum of All Nodes / Check Balanced Tree
Day 57	Trees - Basics	Diameter of Binary Tree / Mirror / Invert Binary Tree
Day 58	Trees - Basics	Path from Root to Node / Lowest Common Ancestor
Day 59	Binary Search Tree (BST)	Search in BST / Insert in BST

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Day 60	Binary Search Tree (BST)	Delete in BST / Kth Smallest in BST

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Month 3 — Overview

Focus: Trees · BST · Heaps · Graphs (BFS, DFS, Dijkstra, MST)

Day	Topic	Questions
Day 61	Binary Search Tree (BST)	Validate BST / Inorder Successor / Predecessor
Day 62	Binary Search Tree (BST)	Convert Sorted Array to BST / BST to Greater Sum Tree
Day 63	Heaps / Priority Queue	Min Heap & Max Heap / Heapify
Day 64	Heaps / Priority Queue	Heap Sort / Kth Largest Element
Day 65	Heaps / Priority Queue	Top K Frequent Elements (Heap) / Merge K Sorted Lists
Day 66	Heaps / Priority Queue, Graphs - Basics	Find Median from Data Stream / Graph Representation (Adjacency List & Matrix)
Day 67	Graphs - Basics	BFS on Graph / DFS on Graph
Day 68	Graphs - Basics	Detect Cycle (Undirected - BFS/DFS) / Detect Cycle (Directed - DFS)
Day 69	Graphs - Basics	Topological Sort (BFS - Kahn's) / Topological Sort (DFS)
Day 70	Graphs - Basics	Bipartite Check / Number of Islands
Day 71	Graphs - Basics, Graphs - Advanced	Flood Fill / Dijkstra's Algorithm
Day 72	Graphs - Advanced	Bellman-Ford Algorithm / Floyd-Warshall Algorithm
Day 73	Graphs - Advanced	Prim's Algorithm (MST) / Kruskal's Algorithm (MST)
Day 74	Graphs - Advanced	Bridges in Graph / Articulation Points
Day 75	Graphs - Advanced, Dynamic Programming - 1D	Strongly Connected Components (Kosaraju) / Fibonacci (Memoization & Tabulation)
Day 76	Dynamic Programming - 1D	Climbing Stairs / House Robber
Day 77	Dynamic Programming - 1D	House Robber II (Circular) / Jump Game
Day 78	Dynamic Programming - 1D	Jump Game II (Min Jumps) / Decode Ways
Day 79	Dynamic Programming - 1D	Minimum Cost Climbing Stairs / Partition Equal Subset Sum
Day 80	Dynamic Programming - 2D (Grid)	Unique Paths / Unique Paths II (Obstacles)
Day 81	Dynamic Programming - 2D (Grid)	Minimum Path Sum / Triangle (Min Path)
Day 82	Dynamic Programming - 2D (Grid)	Maximal Square / Dungeon Game
Day 83	Dynamic Programming - 2D (Grid), DP - Subsequences & Substrings	Cherry Pickup / Longest Common Subsequence (LCS)
Day 84	DP - Subsequences & Substrings	Edit Distance / Longest Increasing Subsequence
Day 85	DP - Subsequences & Substrings	Longest Common Substring / Shortest Common Supersequence
Day 86	DP - Subsequences & Substrings	Count Distinct Subsequences / Wildcard Matching
Day 87	DP - Subsequences & Substrings, DP - Knapsack Variants	Regular Expression Matching / 0/1 Knapsack
Day 88	DP - Knapsack Variants	Unbounded Knapsack / Fractional Knapsack (Greedy)

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Day 89	DP - Knapsack Variants	Rod Cutting / Coin Change (Min Coins)
Day 90	DP - Knapsack Variants	Coin Change II (Count Ways) / Target Sum (DP)

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Month 4 — Overview

Focus: DP · Tries · Segment Trees · Greedy · Sliding Window

Day	Topic	Questions
Day 91	DP - Knapsack Variants, Tries	Boolean Parenthesization / Trie Insert & Search
Day 92	Tries	Prefix Search / Auto-Complete / Count Words with Prefix
Day 93	Tries	Longest Common Prefix (Trie) / Word Break Problem (Trie)
Day 94	Tries, Segment Trees & Advanced DS	Maximum XOR of Two Numbers / Segment Tree - Build
Day 95	Segment Trees & Advanced DS	Segment Tree - Range Query / Segment Tree - Point Update
Day 96	Segment Trees & Advanced DS	Lazy Propagation / Fenwick Tree (BIT)
Day 97	Segment Trees & Advanced DS, Greedy Algorithms	Sparse Table (Range Min/Max) / Activity Selection Problem
Day 98	Greedy Algorithms	Fractional Knapsack / Job Sequencing with Deadlines
Day 99	Greedy Algorithms	Huffman Encoding / N Meetings in One Room
Day 100	Greedy Algorithms	Minimum Platforms / Candy Distribution
Day 101	Greedy Algorithms, Sliding Window & Two Pointers	Gas Station / Maximum Sum Subarray of Size K
Day 102	Sliding Window & Two Pointers	Longest Substring Without Repeating Chars / Minimum Window Substring
Day 103	Sliding Window & Two Pointers	Longest Substring with At Most K Distinct / Count Subarrays with Product Less than K
Day 104	Sliding Window & Two Pointers	Container with Most Water / Trapping Rain Water
Day 105	Sliding Window & Two Pointers	4-Sum Problem

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□ Week-wise Detailed Plan

Granular week-by-week schedule. Each week covers specific topics to reinforce learning progressively.

Month 1 — Foundations (Weeks 1–4)

Week 1

Day	Questions to Solve
Day 1	Q: Big O Notation Basics [Introduction & Time-Space Complexity] | Q: Time Complexity Analysis [Introduction & Time-Space Complexity]
Day 2	Q: Space Complexity Analysis [Introduction & Time-Space Complexity] | Q: Best / Worst / Average Case [Introduction & Time-Space Complexity]
Day 3	Q: Asymptotic Notations [Introduction & Time-Space Complexity] | Q: Introduction to Arrays [Arrays - Basics]
Day 4	Q: Linear Search [Arrays - Basics] | Q: Binary Search [Arrays - Basics]
Day 5	Q: Bubble Sort [Arrays - Basics] | Q: Selection Sort [Arrays - Basics]
Day 6	Q: Insertion Sort [Arrays - Basics] | Q: Find Max/Min in Array [Arrays - Basics]
Day 7	Q: Reverse an Array [Arrays - Basics] | Q: Check Sorted Array [Arrays - Basics]

Week 2

Day	Questions to Solve
Day 8	Q: Find Duplicate Element [Arrays - Intermediate] | Q: Rotate Array by K [Arrays - Intermediate]
Day 9	Q: Move Zeros to End [Arrays - Intermediate] | Q: Two Sum (Unsorted) [Arrays - Intermediate]
Day 10	Q: Two Sum (Sorted) [Arrays - Intermediate] | Q: Pair Sum Closest to Target [Arrays - Intermediate]
Day 11	Q: Triplet Sum [Arrays - Intermediate] | Q: Subarray with Zero Sum [Arrays - Intermediate]
Day 12	Q: Subarray Sum Equals K [Arrays - Intermediate] | Q: Factorial using Recursion [Recursion & Backtracking]
Day 13	Q: Fibonacci using Recursion [Recursion & Backtracking] | Q: Power(x, n) [Recursion & Backtracking]
Day 14	Q: Print Numbers 1 to N [Recursion & Backtracking] | Q: Reverse a String using Recursion [Recursion & Backtracking]

Week 3

Day	Questions to Solve
Day 15	Q: Check Palindrome using Recursion [Recursion & Backtracking] | Q: Tower of Hanoi [Recursion & Backtracking]
Day 16	Q: Print All Subsequences [Recursion & Backtracking] | Q: Print All Permutations of String [Recursion & Backtracking]

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Day 17	Q: N-Queens Problem [Recursion & Backtracking] | Q: Rat in a Maze [Recursion & Backtracking]
Day 18	Q: Sudoku Solver [Recursion & Backtracking] | Q: Reverse a String [Strings]
Day 19	Q: Check Palindrome [Strings] | Q: Count Vowels & Consonants [Strings]
Day 20	Q: Remove Duplicates from String [Strings] | Q: Anagram Check [Strings]
Day 21	Q: Longest Common Prefix [Strings] | Q: Count and Say [Strings]

Week 4

Day	Questions to Solve
Day 22	Q: Roman to Integer [Strings] | Q: Integer to Roman [Strings]
Day 23	Q: Valid Parentheses [Strings] | Q: Longest Palindromic Substring (Brute Force) [Strings]
Day 24	Q: Check Prime Number [Maths for DSA] | Q: Sieve of Eratosthenes [Maths for DSA]
Day 25	Q: GCD & LCM [Maths for DSA] | Q: Count Digits [Maths for DSA]
Day 26	Q: Reverse a Number [Maths for DSA] | Q: Check Armstrong Number [Maths for DSA]
Day 27	Q: Sum of Digits [Maths for DSA] | Q: Count Set Bits [Maths for DSA]
Day 28	Q: Power of 2 [Maths for DSA] | Q: Check if Kth Bit Set [Bit Manipulation]

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Month 2 — Linear DS & Hashing (Weeks 5–8)

Week 5

Day	Questions to Solve
Day 29	Q: Set Kth Bit [Bit Manipulation] | Q: Clear Kth Bit [Bit Manipulation]
Day 30	Q: Toggle Kth Bit [Bit Manipulation] | Q: Count Set Bits (Brian Kernighan) [Bit Manipulation]
Day 31	Q: Find Single Non-Duplicate [Bit Manipulation] | Q: XOR of All Subsets [Bit Manipulation]
Day 32	Q: Introduction & Node Creation [Linked List] | Q: Insert at Beginning / End / Position [Linked List]
Day 33	Q: Delete at Beginning / End / Position [Linked List] | Q: Search in Linked List [Linked List]
Day 34	Q: Reverse a Linked List [Linked List] | Q: Detect Loop (Floyd's Algorithm) [Linked List]
Day 35	Q: Remove Loop [Linked List] | Q: Find Middle Element [Linked List]

Week 6

Day	Questions to Solve
Day 36	Q: Merge Two Sorted Lists [Linked List] | Q: Check Palindrome Linked List [Linked List]
Day 37	Q: Intersection Point of Two Lists [Linked List] | Q: Stack using Array [Stack]
Day 38	Q: Stack using Linked List [Stack] | Q: Valid Parentheses (Stack) [Stack]
Day 39	Q: Next Greater Element [Stack] | Q: Next Smaller Element [Stack]
Day 40	Q: Stock Span Problem [Stack] | Q: Min Stack [Stack]
Day 41	Q: Evaluate Postfix Expression [Stack] | Q: Infix to Postfix [Stack]
Day 42	Q: Queue using Array [Queue] | Q: Queue using Linked List [Queue]

Week 7

Day	Questions to Solve
Day 43	Q: Queue using Two Stacks [Queue] | Q: Circular Queue [Queue]
Day 44	Q: Deque (Double Ended Queue) [Queue] | Q: Sliding Window Maximum [Queue]
Day 45	Q: First Non-Repeating Character in Stream [Queue] | Q: Search in Rotated Sorted Array [Binary Search (Advanced)]
Day 46	Q: Find Peak Element [Binary Search (Advanced)] | Q: Median of Two Sorted Arrays (Brute) [Binary Search (Advanced)]
Day 47	Q: Kth Smallest / Largest [Binary Search (Advanced)] | Q: Allocate Books (Binary Search on Answer) [Binary Search (Advanced)]
Day 48	Q: Painter's Partition [Binary Search (Advanced)] | Q: Aggressive Cows [Binary Search (Advanced)]
Day 49	Q: Capacity to Ship Packages [Binary Search (Advanced)] | Q: HashMap Basics [Hashing]

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Week 8

Day	Questions to Solve
Day 50	Q: Frequency Count [Hashing] | Q: Two Sum using Hashing [Hashing]
Day 51	Q: Longest Consecutive Sequence [Hashing] | Q: Group Anagrams [Hashing]
Day 52	Q: Subarray Sum Equals K (Hashing) [Hashing] | Q: Top K Frequent Elements [Hashing]
Day 53	Q: Intersection & Union of Arrays [Hashing] | Q: Binary Tree Introduction & Node Creation [Trees - Basics]
Day 54	Q: Tree Traversals (Inorder, Preorder, Postorder) [Trees - Basics] | Q: Level Order Traversal (BFS) [Trees - Basics]
Day 55	Q: Height of Binary Tree [Trees - Basics] | Q: Count Nodes [Trees - Basics]
Day 56	Q: Sum of All Nodes [Trees - Basics] | Q: Check Balanced Tree [Trees - Basics]

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Month 3 — Trees, Graphs & BST (Weeks 9–12)

Week 9

Day	Questions to Solve
Day 57	Q: Diameter of Binary Tree [Trees - Basics] | Q: Mirror / Invert Binary Tree [Trees - Basics]
Day 58	Q: Path from Root to Node [Trees - Basics] | Q: Lowest Common Ancestor [Trees - Basics]
Day 59	Q: Search in BST [Binary Search Tree (BST)] | Q: Insert in BST [Binary Search Tree (BST)]
Day 60	Q: Delete in BST [Binary Search Tree (BST)] | Q: Kth Smallest in BST [Binary Search Tree (BST)]
Day 61	Q: Validate BST [Binary Search Tree (BST)] | Q: Inorder Successor / Predecessor [Binary Search Tree (BST)]
Day 62	Q: Convert Sorted Array to BST [Binary Search Tree (BST)] | Q: BST to Greater Sum Tree [Binary Search Tree (BST)]
Day 63	Q: Min Heap & Max Heap [Heaps / Priority Queue] | Q: Heapify [Heaps / Priority Queue]

Week 10

Day	Questions to Solve
Day 64	Q: Heap Sort [Heaps / Priority Queue] | Q: Kth Largest Element [Heaps / Priority Queue]
Day 65	Q: Top K Frequent Elements (Heap) [Heaps / Priority Queue] | Q: Merge K Sorted Lists [Heaps / Priority Queue]
Day 66	Q: Find Median from Data Stream [Heaps / Priority Queue] | Q: Graph Representation (Adjacency List & Matrix) [Graphs - Basics]
Day 67	Q: BFS on Graph [Graphs - Basics] | Q: DFS on Graph [Graphs - Basics]
Day 68	Q: Detect Cycle (Undirected - BFS/DFS) [Graphs - Basics] | Q: Detect Cycle (Directed - DFS) [Graphs - Basics]
Day 69	Q: Topological Sort (BFS - Kahn's) [Graphs - Basics] | Q: Topological Sort (DFS) [Graphs - Basics]
Day 70	Q: Bipartite Check [Graphs - Basics] | Q: Number of Islands [Graphs - Basics]

Week 11

Day	Questions to Solve
Day 71	Q: Flood Fill [Graphs - Basics] | Q: Dijkstra's Algorithm [Graphs - Advanced]
Day 72	Q: Bellman-Ford Algorithm [Graphs - Advanced] | Q: Floyd-Warshall Algorithm [Graphs - Advanced]
Day 73	Q: Prim's Algorithm (MST) [Graphs - Advanced] | Q: Kruskal's Algorithm (MST) [Graphs - Advanced]
Day 74	Q: Bridges in Graph [Graphs - Advanced] | Q: Articulation Points [Graphs - Advanced]
Day 75	Q: Strongly Connected Components (Kosaraju) [Graphs - Advanced] | Q: Fibonacci (Memoization & Tabulation) [Dynamic Programming - 1D]
Day 76	Q: Climbing Stairs [Dynamic Programming - 1D] | Q: House Robber [Dynamic Programming - 1D]

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Day 77	Q: House Robber II (Circular) [Dynamic Programming - 1D] | Q: Jump Game [Dynamic Programming - 1D]

Week 12

Day	Questions to Solve
Day 78	Q: Jump Game II (Min Jumps) [Dynamic Programming - 1D] | Q: Decode Ways [Dynamic Programming - 1D]
Day 79	Q: Minimum Cost Climbing Stairs [Dynamic Programming - 1D] | Q: Partition Equal Subset Sum [Dynamic Programming - 1D]
Day 80	Q: Unique Paths [Dynamic Programming - 2D (Grid)] | Q: Unique Paths II (Obstacles) [Dynamic Programming - 2D (Grid)]
Day 81	Q: Minimum Path Sum [Dynamic Programming - 2D (Grid)] | Q: Triangle (Min Path) [Dynamic Programming - 2D (Grid)]
Day 82	Q: Maximal Square [Dynamic Programming - 2D (Grid)] | Q: Dungeon Game [Dynamic Programming - 2D (Grid)]
Day 83	Q: Cherry Pickup [Dynamic Programming - 2D (Grid)] | Q: Longest Common Subsequence (LCS) [DP - Subsequences & Substrings]
Day 84	Q: Edit Distance [DP - Subsequences & Substrings] | Q: Longest Increasing Subsequence [DP - Subsequences & Substrings]

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Month 4 — DP, Greedy & Advanced DS (Weeks 13–16)

Week 13

Day	Questions to Solve
Day 85	Q: Longest Common Substring [DP - Subsequences & Substrings] | Q: Shortest Common Supersequence [DP - Subsequences & Substrings]
Day 86	Q: Count Distinct Subsequences [DP - Subsequences & Substrings] | Q: Wildcard Matching [DP - Subsequences & Substrings]
Day 87	Q: Regular Expression Matching [DP - Subsequences & Substrings] | Q: 0/1 Knapsack [DP - Knapsack Variants]
Day 88	Q: Unbounded Knapsack [DP - Knapsack Variants] | Q: Fractional Knapsack (Greedy) [DP - Knapsack Variants]
Day 89	Q: Rod Cutting [DP - Knapsack Variants] | Q: Coin Change (Min Coins) [DP - Knapsack Variants]
Day 90	Q: Coin Change II (Count Ways) [DP - Knapsack Variants] | Q: Target Sum (DP) [DP - Knapsack Variants]
Day 91	Q: Boolean Parenthesization [DP - Knapsack Variants] | Q: Trie Insert & Search [Tries]

Week 14

Day	Questions to Solve
Day 92	Q: Prefix Search / Auto-Complete [Tries] | Q: Count Words with Prefix [Tries]
Day 93	Q: Longest Common Prefix (Trie) [Tries] | Q: Word Break Problem (Trie) [Tries]
Day 94	Q: Maximum XOR of Two Numbers [Tries] | Q: Segment Tree - Build [Segment Trees & Advanced DS]
Day 95	Q: Segment Tree - Range Query [Segment Trees & Advanced DS] | Q: Segment Tree - Point Update [Segment Trees & Advanced DS]
Day 96	Q: Lazy Propagation [Segment Trees & Advanced DS] | Q: Fenwick Tree (BIT) [Segment Trees & Advanced DS]
Day 97	Q: Sparse Table (Range Min/Max) [Segment Trees & Advanced DS] | Q: Activity Selection Problem [Greedy Algorithms]
Day 98	Q: Fractional Knapsack [Greedy Algorithms] | Q: Job Sequencing with Deadlines [Greedy Algorithms]

Week 15

Day	Questions to Solve
Day 99	Q: Huffman Encoding [Greedy Algorithms] | Q: N Meetings in One Room [Greedy Algorithms]
Day 100	Q: Minimum Platforms [Greedy Algorithms] | Q: Candy Distribution [Greedy Algorithms]
Day 101	Q: Gas Station [Greedy Algorithms] | Q: Maximum Sum Subarray of Size K [Sliding Window & Two Pointers]

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Day 102	Q: Longest Substring Without Repeating Chars [Sliding Window & Two Pointers] | Q: Minimum Window Substring [Sliding Window & Two Pointers]
Day 103	Q: Longest Substring with At Most K Distinct [Sliding Window & Two Pointers] | Q: Count Subarrays with Product Less than K [Sliding Window & Two Pointers]
Day 104	Q: Container with Most Water [Sliding Window & Two Pointers] | Q: Trapping Rain Water [Sliding Window & Two Pointers]
Day 105	Q: 4-Sum Problem [Sliding Window & Two Pointers]

Week 16

Day	Questions to Solve

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□ Study Tips & Strategy

Follow these tips for maximum retention and interview readiness:

- ✅Solve questions first on paper before coding — builds problem-solving intuition.
- ✅Do not skip recursion and backtracking — they are the backbone of DP and graphs.
- ✅Revise previous week's questions every Sunday before starting the new week.
- ✅After solving, check for optimal solutions even if your brute force passes.
- ✅Practice on LeetCode / GFG for each topic as you complete it.
- ✅Month 4 (DP) is the hardest — allocate extra time if needed and don't rush.
- ✅Track your daily progress — consistency beats intensity.
- ✅Take 1 full day off per week (Sunday) to revise and consolidate learning.

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